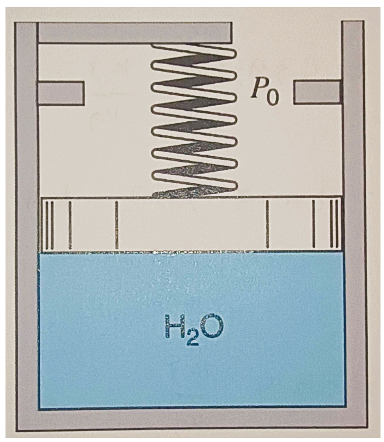
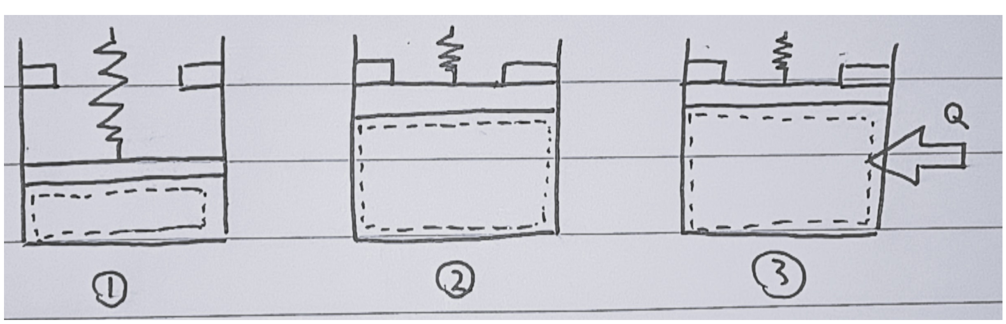
ESC 330 Exam 1 Resub (Problem 2)

Problem 2: (50 points) [Show work and explain your solution for partial credit]:
Two kilograms of **water** is contained in a **piston/cylinder** (see figure) with a **massless piston** loaded with a **linear spring** and the outside atmosphere. **Initially** the **spring force is zero** and **$P_1 = P_0 = 100$ kPa** with a volume of **0.2 m³**. If the piston just hits the upper stops, the volume is **0.8 m³** and **T = 600°C**. Heat is now added until the pressure reaches **1.2 MPa**.
a) [30 Points] Show the **P-V diagram**
b) [10 Points] Determine the **work** for the process, in **KJ**.
c) [10 Points] Determine the **heat transfer** for the process, in **KJ**.



1. **Objective:**
   - Show the P-$\cancel{V}$ diagram for the process.
   - Find the work for the process W [kJ].
   - Find the amount of heat transfer for the process Q [kJ].
2. **Schematic:**



3. **Given:**

   - *Closed system*: containing $m = 2$ kg of *water*.
   - State 1: $P_1 = 100$ kPa, $V_1 = 0.2$ m^3 $\implies$ $v_1 = 0.1$ m^3/kg
   - State 2: $V_2 = 0.8$ m^3, $T_2 = 600$°C $\implies$ $v_2 = 0.4$ m^3/kg
   - State 3: $V_3 = 0.8$ m^3, $P_3 = 1.2$ MPa $\implies$ $v_3 = 0.4$ m^3/kg

4. **Approach:**

   - Use property tables to determine phases and properties for states 1-3 and plot them onto a P-V diagram.
   - Determine work of the process using the equation of work for a *closed system*:

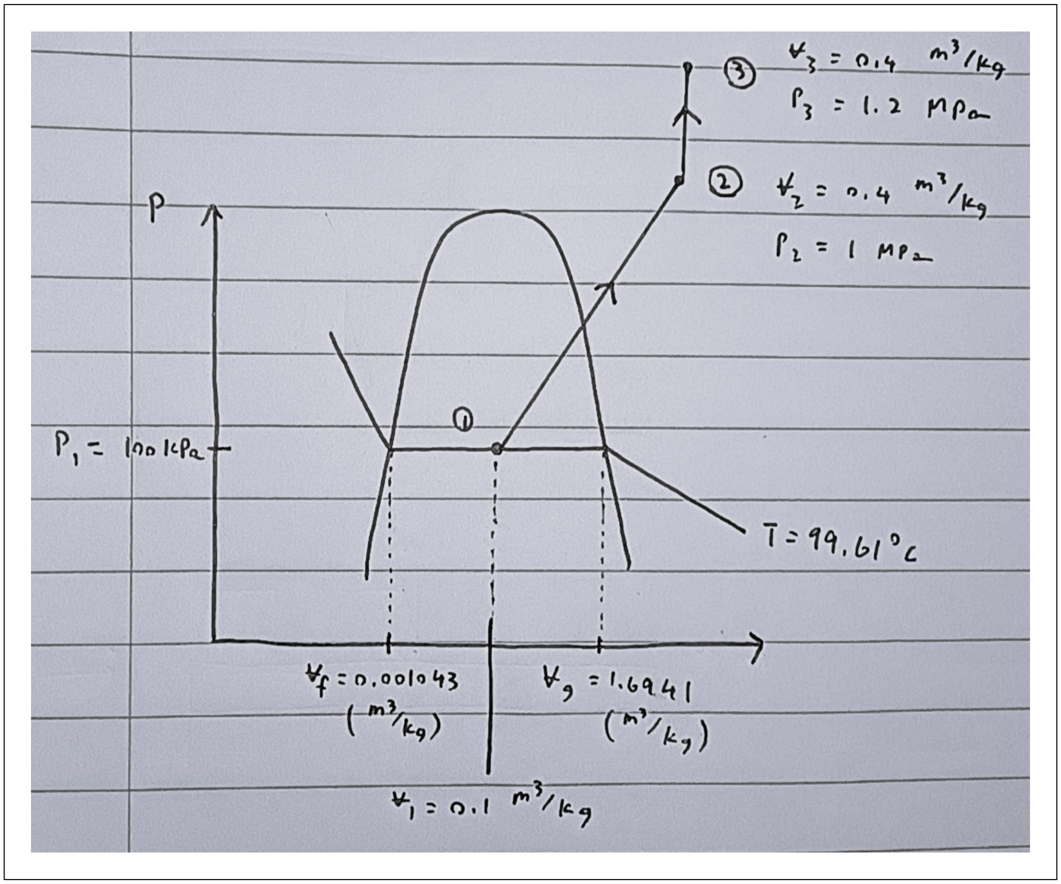
   $$W_{\text{closed}} = \int P\, dV = m \int P\, dv$$

   - Apply the first law for a closed system to determine the heat transfer for the process Q:

   $$\Delta U = Q - W$$

5. **P-V Diagram:**

   - State 1: at $P_1 = 100$ kPa, $v_f < v_1 < v_g$ implies that state 1 is a *saturated mixture*. Thus, its temperature is the saturation temperature at P_1, which is $T_1 = 99.61$°C.
   - State 2: at $T_2 = 600$°C, it is a *superheated vapor*. Looking at the superheated water tables, we find that for $T_2 = 600$°C and $v_2 = 0.40111 \approx 0.4$ m^3/kg, the pressure is $P_2 = 1$ MPa.
   - State 3: Since state 2 is a superheated vapor and heat is added from state 2 to state 3, state 3 must also be a superheated vapor. Since volume is constant from state 2 to state 3, we have $v_3 = v_2 = 0.4$ m^3/kg (isochoric process). We are given that $P_3 = 1.2$ MPa.



6. **Work of a closed system:**
Since there is no change of volume from state 2 to state 3, there is no work done during this process. Thus, we only need to calculate the work done from state 1 to state 2:

$$W_{1\to 2} = m \int_{v_1}^{v_2} P \, dv$$

Geometrically, this is the area under the curve from state 1 to state 2 on the P-V diagram. This is the same as the area of the trapezoid formed by states 1 and 2:

$$\begin{aligned} W_{1\to 2} &= m \left(\frac{P_1 + P_2}{2} \right) (v_2 - v_1) \\ &= [2\,\text{kg}] \left(\frac{[100 + 1000\,\text{kPa}]}{2} \right) ([0.4 - 0.1\,\text{m}^3/\text{kg}]) \\ &= 330\,\text{kJ} \end{aligned}$$

$$\boxed{\implies W_{1\to 3} = W_{1\to 2} + \underbrace{\cancel{W_{2\to 3}}}_{0} = 330\,\text{kJ}}$$

Note: A quicker way to approximate the work is to use a rectangular area using the average pressure $P_{\text{avg}} = (P_1 + P_2)/2$ and the change in volume $\Delta V = m(v_2 - v_1)$.

7. **Heat transfer:**
Knowing work, we just need to use the property tables to find the specific internal energies at states 1 and 3:

- State 1:
Since state 1 is a saturated mixture, we can find the quality x_1 using the specific volume:
$$x_1 = \frac{v_1 - v_f}{v_g - v_f} = \frac{[0.1 - 0.001043\,\mathrm{m^3/kg}]}{[1.6941 - 0.001043\,\mathrm{m^3/kg}]} = 0.0584$$
Use quality to find the specific internal energy at state 1:
$$\begin{aligned} u_1 &= u_f + x_1 u_{fg} \\ &= [417.40\,\mathrm{kJ/kg}] + 0.0584[2088.2\,\mathrm{kJ/kg}] = 539.45\,\mathrm{kJ/kg} \end{aligned}$$

- State 3:
Linear interpolation for the superheated water tables at $P_3 = 1.2$ MPa, using the values at
  - $T = 700°\mathrm{C}$: $v = 0.37297$ m^3/kg, $u = 3475.3$ kJ/kg
  - $T = 800°\mathrm{C}$: $v = 0.41184$ m^3/kg, $u = 3661.0$ kJ/kg
$$\begin{aligned} u - u_{800} &= \frac{u_{800} - u_{700}}{v_{800} - v_{700}}(v - v_{800}) \\ u &= 3661.0 + 4777.46(v - 0.41184) \\ &\implies u_3 = 3604.43\,\mathrm{kJ/kg} \end{aligned}$$

Apply the first law for a closed system to find the heat transfer:
$$\begin{aligned} \Delta U &= Q - W \\ m(u_3 - u_1) &= Q - W \\ Q &= m(u_3 - u_1) + W \end{aligned}$$
$$Q = [2\,\mathrm{kg}]([3604.43 - 539.45\,\mathrm{kJ/kg}]) + [330\,\mathrm{kJ}] = \boxed{6459.96\,\mathrm{kJ}}$$